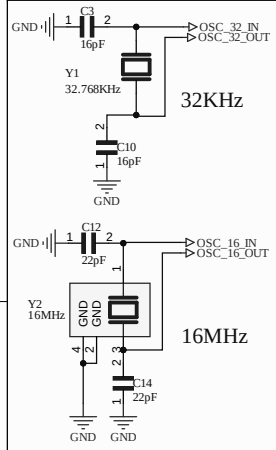
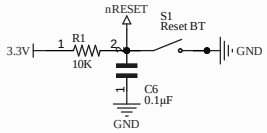
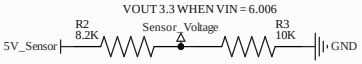
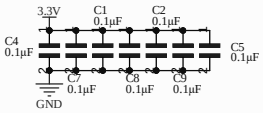
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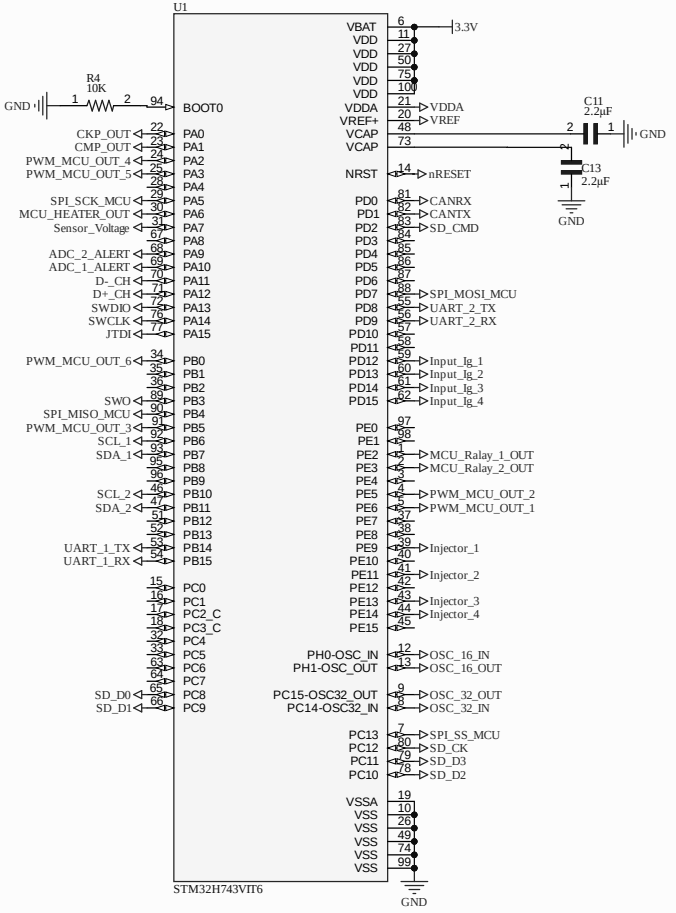
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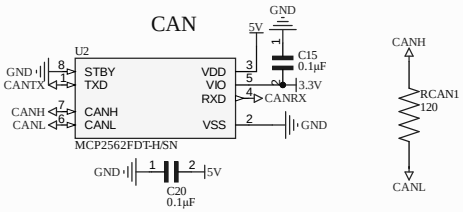
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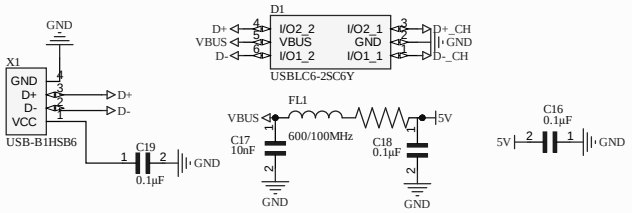
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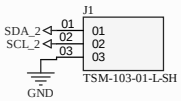
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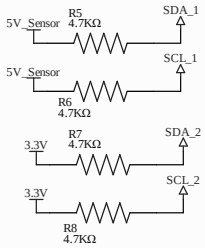
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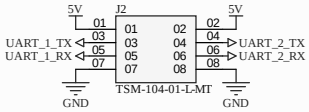


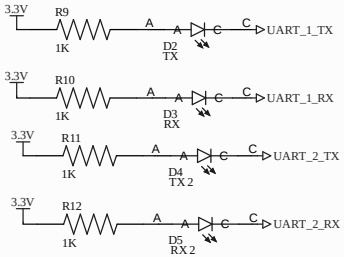
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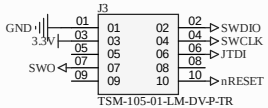


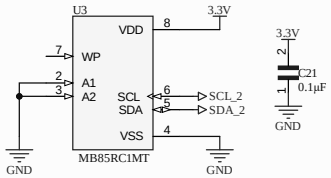


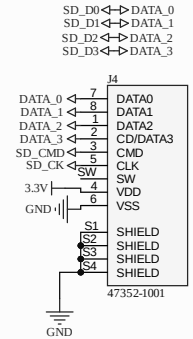
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PROJECT: Altium

PROJECT REVISION: 24082465779156992979f20d2a4d08e8e118 DOCUMENT REVISION: DESIGN ITEM:

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SCALE: FILE NAME: CommunicationInterfaces.SchDoc SHEET 2 OF 12

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REVISION	DESCRIPTION	DATE	APPROVED

VOUT 3.3 WHEN VIN = 18.81

R16 47K, Bat_Voltage, R17 10K, VIN, GND

U4 PC817X3NSZ9F: VIN, R14 1.2K, Neutral_Gear_Sensor, GND, R15 10K, R13 1K, Neutral_Gear_Sensor_OUT, C22 0.1µF

U7 MIC2981/82YWM: VS, VIN, GND
PWM_VOLTAGE_SELEC_1..4 → IN1..IN4 → OUT1..OUT4 → PWM_12V_IN_1..4
IN5..IN8, OUT5..OUT8

D6..D9 BAT54: 5V → PWM_12V_IN_1..4 → R18..R21 4.7K → PWM_OUT_1..4

U6 ADS8688IDBTR

Pin	Net
AVDD, AVDD	5V_Sensor
DVDD	3.3V
RST/PD	GND
REFSEL	GND
DAISY	
REFCAP	REFCAP_1
REFIO	REFIO_1
SDI	SPI_MOSI_ADC
SDO	SPI_MISO_ADC
SCLK	SPI_SCK_ADC
CS	ADC_1_CS
AUX_IN	
AUX_GND	
AIN_0P	Sensor_1
AIN_0GND	GND
AIN_1P	Sensor_2
AIN_1GND	GND
AIN_2P	Sensor_3
AIN_2GND	GND
AIN_3P	Sensor_4
AIN_3GND	GND
AIN_4P	Sensor_5
AIN_4GND	GND
AIN_5P	Sensor_6
AIN_5GND	GND
AIN_6P	CJ125_UA
AIN_6GND	GND
AIN_7P	CJ125_UR
AIN_7GND	GND
DNC	ADC_1_ALERT
AGND ×6	GND
DGND	GND
REFGND	GND

REFCAP_1: C28 22µF, C29 1µF; REFIO_1: C30 10µF; ADC_1_ALERT → R22 10K → 3.3V

5V_Sensor → C24 10µF → GND

U5 ADS8688IDBTR

Pin	Net
AVDD, AVDD	5V_Sensor
DVDD	3.3V
RST/PD	GND
REFSEL	GND
DAISY	
REFCAP	REFCAP_2
REFIO	REFIO_2
SDI	SPI_MOSI_ADC
SDO	SPI_MISO_ADC
SCLK	SPI_SCK_ADC
CS	ADC_2_CS
AUX_IN	
AUX_GND	
AIN_0P	Bat_Voltage
AIN_0GND	GND
AIN_1P	MAP_External_Sensor_In
AIN_1GND	GND
AIN_2P	MAF_External_Sensor_In
AIN_2GND	GND
AIN_3P	Fuel_Pressure_Sensor_In
AIN_3GND	GND
AIN_4P	Oil_Pressure_Sensor_In
AIN_4GND	GND
AIN_5P	TPS_1_Sensor_In
AIN_5GND	GND
AIN_6P	TPS_2_Sensor_In
AIN_6GND	GND
AIN_7P	Accelerator_Pedal_Sensor_In_1
AIN_7GND	GND
DNC	ADC_2_ALERT
AGND ×6	GND
DGND	GND
REFGND	GND

5V_Sensor → C23 10µF → GND

REFCAP_2: C25 22µF, C26 1µF; REFIO_2: C27 10µF; ADC_2_ALERT → R23 10K → 3.3V

TPS_1_Sensor → TVS_16 → TPS_1_Sensor_In → C31 0.1µF → GND; R24 4.7K → 5V_Sensor

TPS_2_Sensor → TVS_17 → TPS_2_Sensor_In → C32 0.1µF → GND; R25 4.7K → 5V_Sensor

Oil_Pressure_Sensor → TVS_12 → Oil_Pressure_Sensor_In → C33 0.1µF → GND; R26 4.7K → 5V_Sensor

Fuel_Pressure_Sensor → TVS_9 → Fuel_Pressure_Sensor_In → C34 0.1µF → GND; R27 4.7K → 5V_Sensor

PWM_OUT_1 → Q1 BSS138, 3.3V, R28 4.7K → PWM_MCU_OUT_1

PWM_OUT_2 → Q2 BSS138, 3.3V, R29 4.7K → PWM_MCU_OUT_2

PWM_OUT_3 → Q3 BSS138, 3.3V, R33 4.7K → PWM_MCU_OUT_3

PWM_OUT_4 → Q4 BSS138, 3.3V, R37 4.7K → PWM_MCU_OUT_4

Sensor_1 → TVS_1 → Sensor_In_1 → C37 0.1µF → GND; R31 4.7K → 5V_Sensor

Sensor_2 → TVS_2 → Sensor_In_2 → C38 0.1µF → GND; R32 4.7K → 5V_Sensor

Sensor_3 → TVS_3 → Sensor_In_3 → C35 0.1µF → GND; R30 4.7K → 5V_Sensor

Sensor_4 → TVS_11 → Sensor_In_4 → C41 0.1µF → GND; R35 4.7K → 5V_Sensor

Sensor_5 → TVS_10 → Sensor_In_5 → C42 0.1µF → GND; R36 4.7K → 5V_Sensor

Sensor_6 → TVS_4 → Sensor_In_6 → C40 0.1µF → GND; R34 4.7K → 5V_Sensor

U8 SP721ABTG: 5V_Sensor → V+; C39 0.1µF → GND; V- → GND

Pin	Net
IN_2	TVS_7
IN_3	TVS_8
IN_4	TVS_3
IN_5	TVS_9
IN_6	TVS_2
IN	TVS_1

U9 SP721ABTG: 5V_Sensor → V+; C36 0.1µF → GND; V- → GND

Pin	Net
IN_2	TVS_18
IN_3	TVS_17
IN_4	TVS_16
IN_5	TVS_15
IN_6	TVS_14
IN	TVS_13

U10 SP721ABTG: 5V_Sensor → V+; C43 0.1µF → GND; V- → GND

Pin	Net
	TVS_5 → CJ125_UA
	TVS_6 → CJ125_UR
IN_2	TVS_6
IN_3	TVS_5
IN_4	TVS_10
IN_5	TVS_4
IN_6	TVS_11
IN	TVS_12

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PROJECT: Altium

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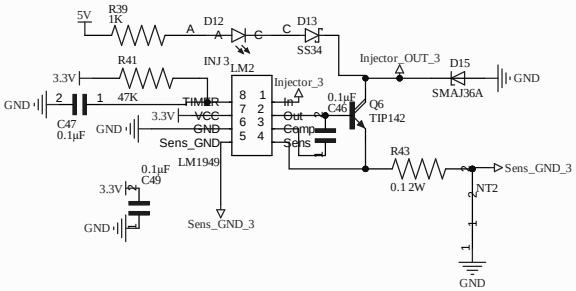
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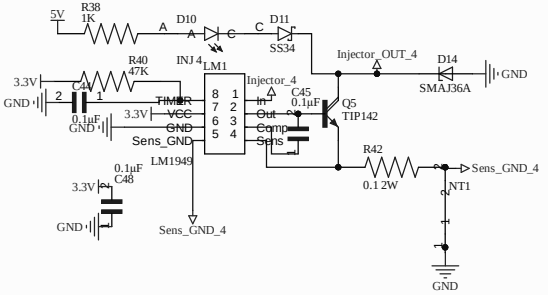
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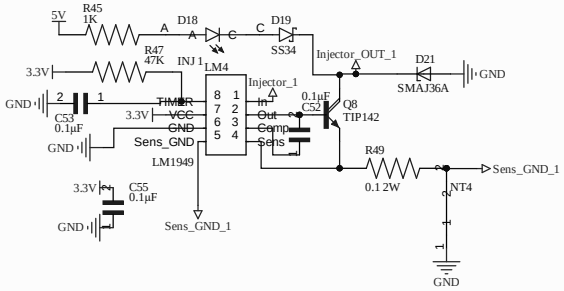
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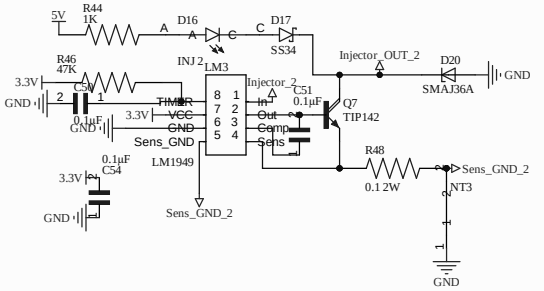
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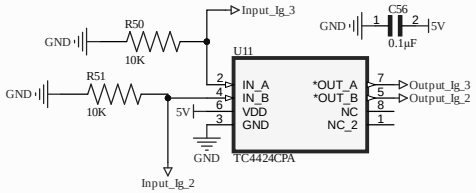
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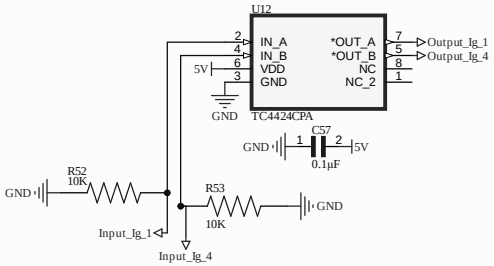












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PROJECT: Altium

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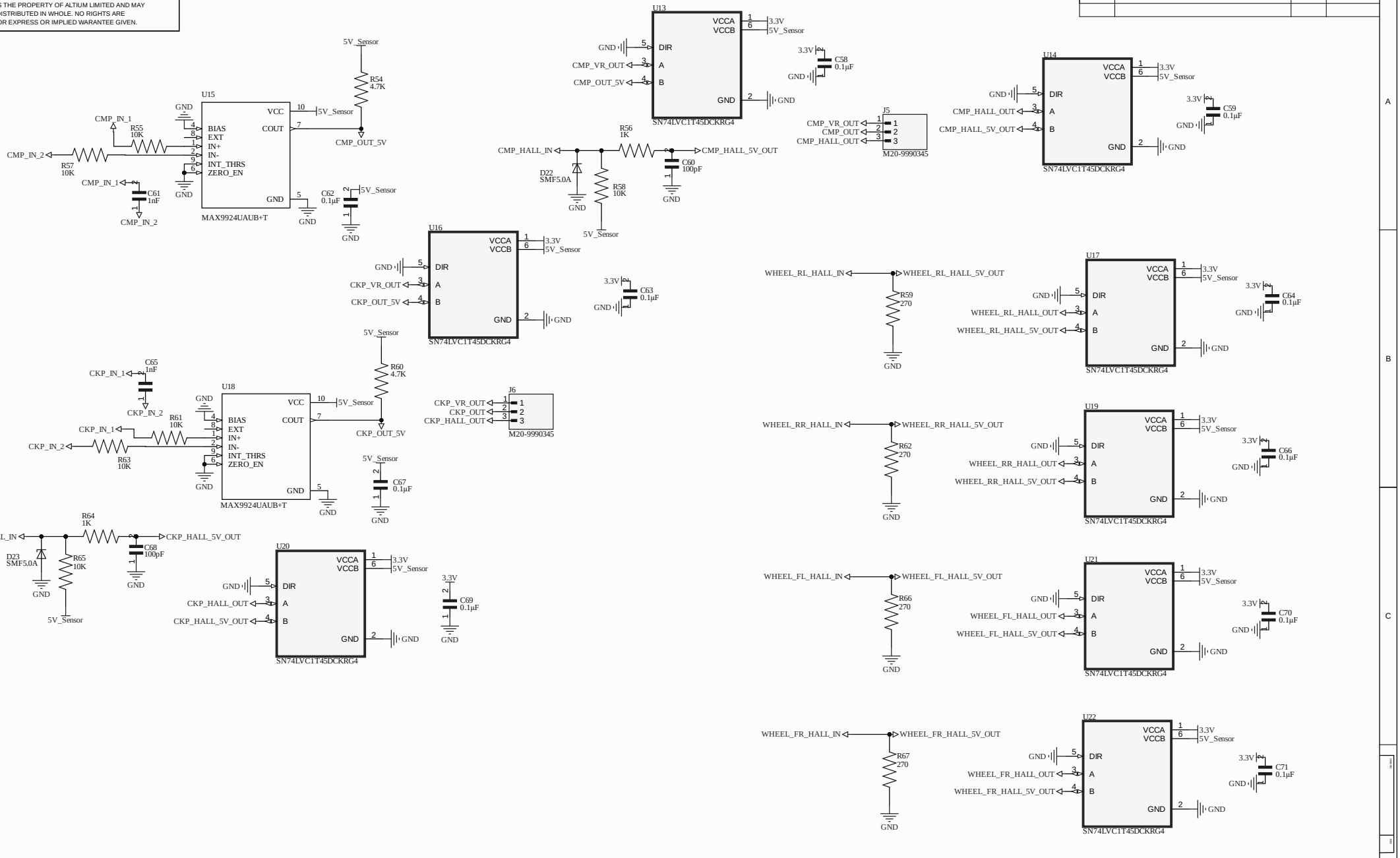
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PROJECT: Altium

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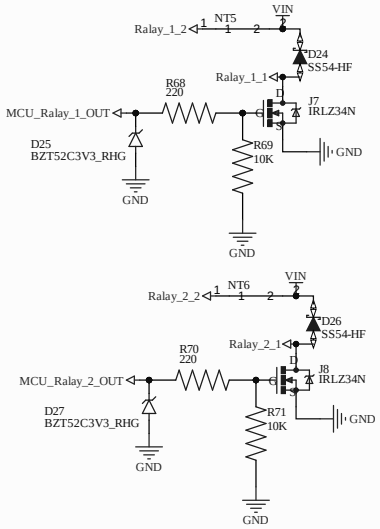
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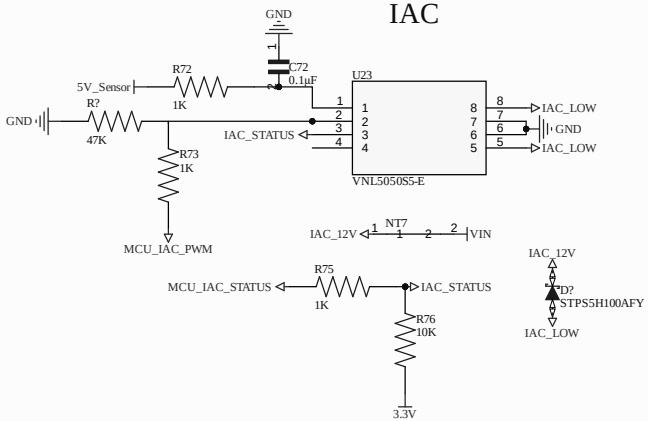
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PROJECT

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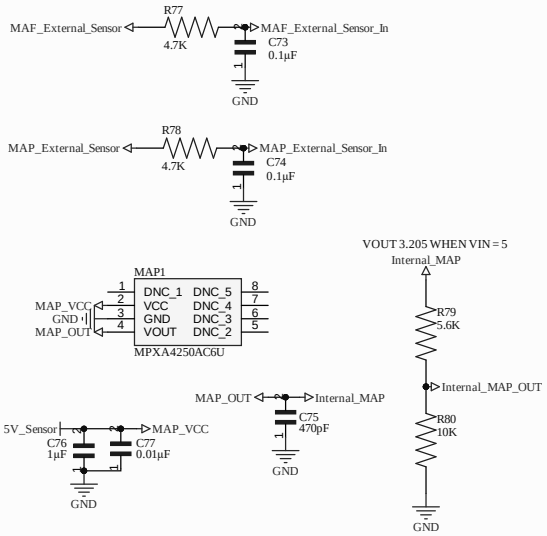
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SCALE: FILE NAME LowSideSwitches.SchDoc SHEET 6 OF 12

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PROJECT: Altium

PROJECT REVISION: 24082465779156995279ff20ck2a4d08ck8e118

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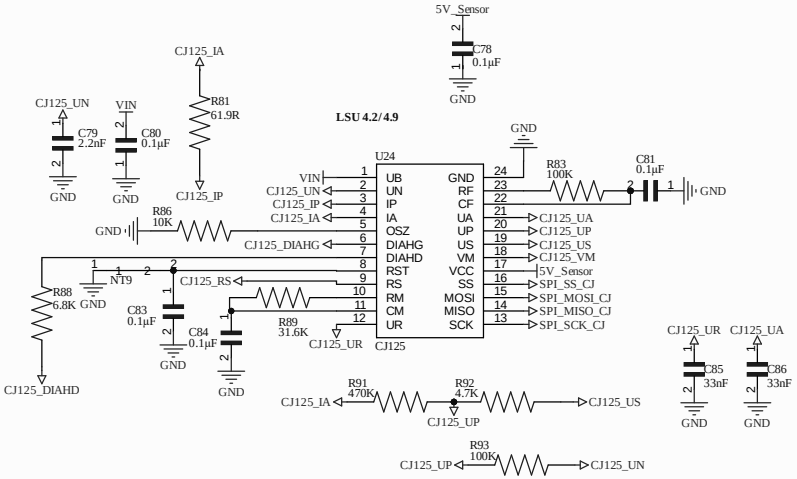
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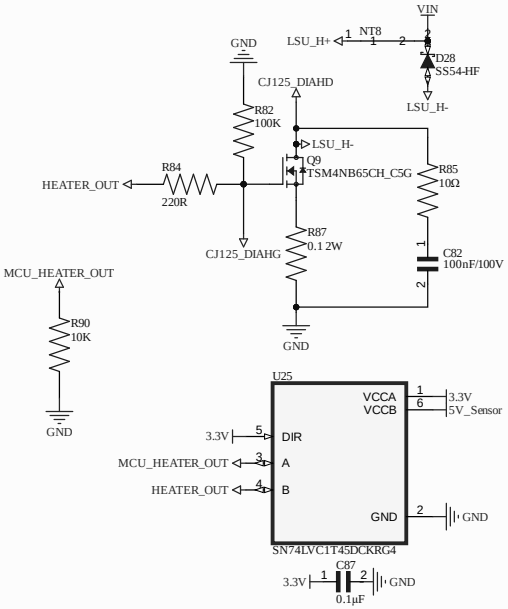
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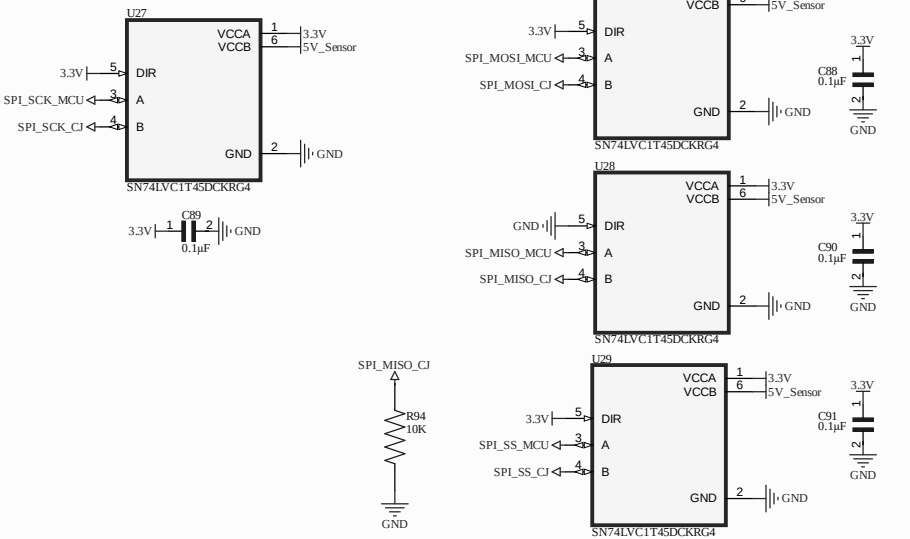
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CJ125_IA ◁▷ LAMBDA_IA
CJ125_UN ◁▷ LAMBDA_UN
CJ125_VM ◁▷ LAMBDA_VM
LSU_H+ ◁▷ LAMBDA_H+
LSU_H- ◁▷ LAMBDA_H-



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PROJECT: Altium

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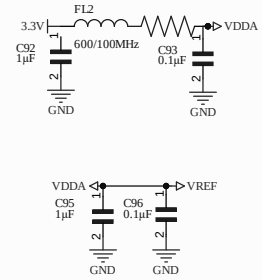
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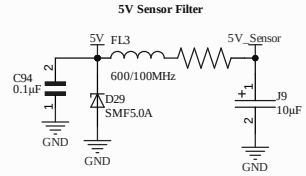
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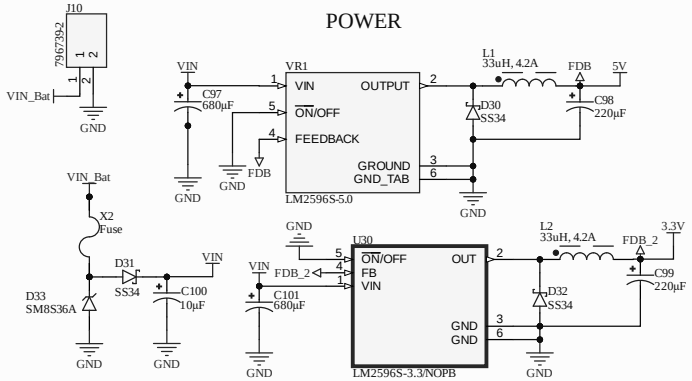
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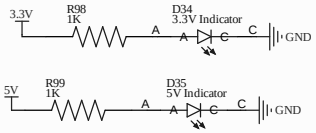
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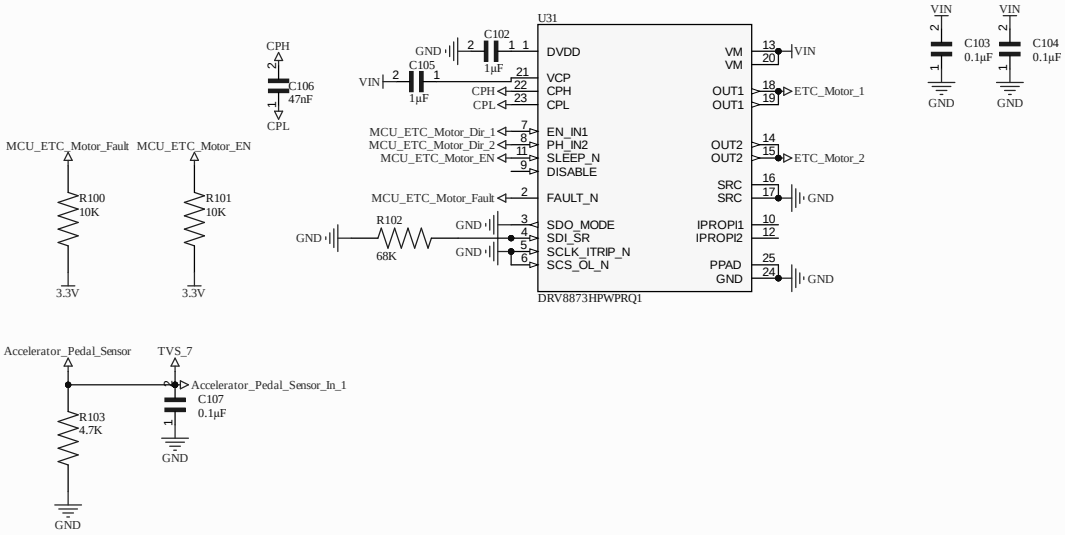
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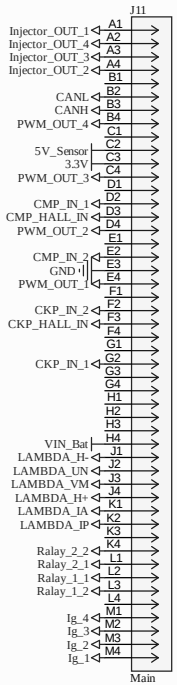
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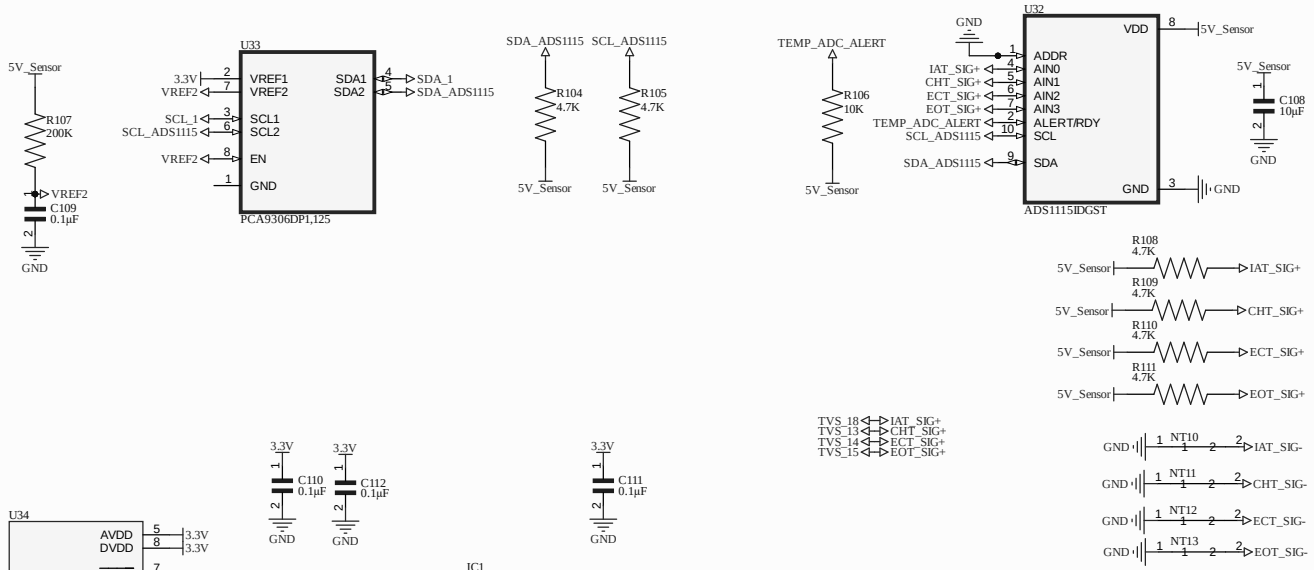
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